

Interpretability and Explainability in Generative Artificial Intelligence

Literature Review

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The popularity of Generative artificial intelligence (GenAI) has recently taken off by storm, with GenAI being incorporated into a diverse variety of industries, including banking, healthcare, autonomous system, and IoT. GenAI has proven useful in applications such as media generation, copywriting, code generation, and conversational/writing assistants, as it dramatically enhances efficiency and speed, personalization, creativity, data analysis/synthesis, design, and content creation, to name only a few of its attractive benefits.

GenAI allows users to input diverse prompts in natural language and generate new artifacts, as opposed to decisions which discriminative models produce. GenAI models utilize neural networks to identify patterns in pre-existing, unlabeled data, and are trained via unsupervised and semi-supervised learning approaches. They are powered by foundation models such as LLMs, GANs, VAEs, Multimodal, and Stable Diffusion.

Unfortunately, due to the opaque, complex algorithms behind ML and DL models, it is challenging to understand their underlying structure to comprehend how they work, making them black-box. Catastrophic failures and unwanted biases in the algorithms can therefore result because humans cannot predict how the algorithms will react to novel inputs. AI powered systems are thus at risk of security attacks, which can leave humans, machines, robots, and data defenseless.

Consequently, there are several ethical and social concerns associated with GenAI models, including “lack of interpretability, bias and discrimination, privacy, lack of model robustness, fake and misleading content, copyright implications, plagiarism, and environmental impact associated with training and inference of generative AI models”. [1]

For example, although LLMs have demonstrated strong reasoning ability and inspired many previously unimaginable applications, they are rooted in the concept of emergent capabilities in which machines statistically learn complex patterns from code data. [2] Therefore, LLMs are susceptible to prompt injection attacks where the model is manipulated to perform unintended actions or generate malicious content [3]. This vulnerability is a significant concern as the adoption of applications that integrate LLMs grows globally.

Furthermore, the integrity of information ecosystems worldwide is profoundly threatened due to proliferation of deepfakes and the spread of misinformation caused by generative AI based on large models that craft seemingly convincing yet fabricated content. [4]

Therefore, when deploying LLMs and other GenAI models, there is a critical need to adapt AI model governance and responsible AI practices to ensure there are robust, comprehensive, and multi-faceted defense strategies implemented that evolve in parallel to the advancement of frontier AI, to protect these cutting-edge yet vulnerable models from attacks.

This is where model interpretability and explainability can have a significant impact.

Interpretability refers to the extent to which a cause and effect can be observed in a system, meaning users can understand a model's reasoning. In contrast, explainability is the ability to describe the internal mechanics, parameters, and architecture of a system, meaning users comprehend how and why a particular output was generated given a certain input.

Therefore, interpretability is critical for building trust in and debugging a model, allowing users to verify that the model is making decisions for the correct reasons, whereas explainability is crucial for improving and refining models, providing insights into how the model works to guide users in improving its performance.

Unfortunately, due to the complex, non-linear nature and high dimensionality of the parameter space of deep generative models, achieving both interpretability and explainability is challenging.

To further complicate matters, the interpretability of AI models is often inversely correlated with their reliability and predictive accuracy. This is demonstrated in the diagram from DARPA XAI research showcasing that although the highest explainability levels are achieved by support vector machines, ensemble models, and decision trees, they have the lowest predictive performance. In contrast, for deep neural networks, the opposite is true. [5]

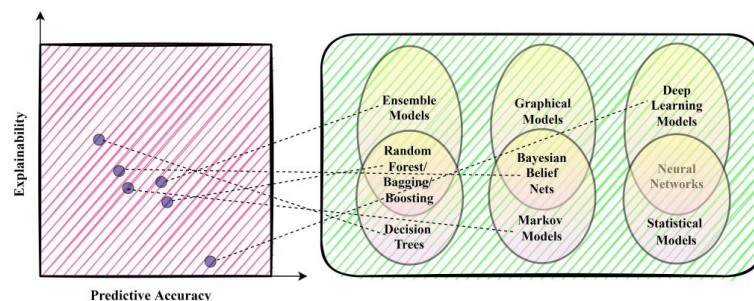


FIGURE 9. Machine learning models' explainability tends to be inversely correlated with their predictive performance.

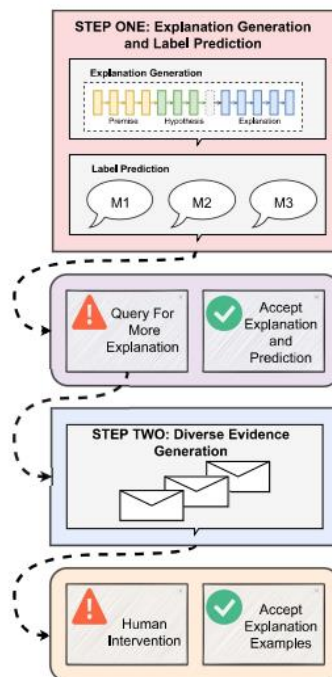
[5]

Currently, model-based, post-hoc, and interactive explainability approaches are employed in practice, achieved through methods including feature attribution, rule extraction, counterfactual analysis, perturbation-based, backpropagation-based, and gradient-based saliency maps, and feature inversion methods. However, XAI faces challenges such as a lack of uniform measurements and evaluation procedures, interpretable data, and domain-specific expertise, which gives rise to ethical and social consequences of XAI, including bias, accountability, and transparency. [6]

Therefore, there has been extensive research conducted in attempts to generate interpretable GenAI models that do not sacrifice performance, reliability, and accuracy.

For example, to mitigate risks stemming from AI-Generated Content, one study proposed a framework integrating advanced detection algorithms, cross-platform collaboration, and policy-driven initiatives, that leveraged multi-modal analysis, digital watermarking, and machine learning-based authentication techniques, as a defense mechanism from deepfakes, misinformation, and disinformation. [4]

INTERACTION (explain aNd predicT thEn queRy with contextuAl CondiTional varlational autO-eNcoder) was a generative XAI framework proposed in another study, which presents model explanations in two steps: Explanation and Label Prediction, and Diverse Evidence Generation. Experiments using the Transformer architecture demonstrated that this two-step approach helped account for the diversity of human thoughts in language in which traditional XAI with natural language processing approaches that deliver a single explanation, fail to account for. [7]



Graphical overview of our framework, **INTERACTION**[7]

Sampling - Variational Auto Encoder (VAE) - Ensemble Anomaly Detection (SVEAD), was a hybrid architecture proposed in another study, that attempts to understand the outputs of AI models. SVEAD compresses data using VAE to a lower-dimensional space, then combines outputs of various anomaly detection models, leveraging ensemble techniques, to identify anomalies in complex datasets. By combining VAE with ensemble stacking and SHapley Additive exPlanations for imbalanced classification, SVEAD proved to lead to superior model performance and an easily explainable framework, as model interpretability is achieved using SHAP combined with Permutation Importance and Individual Conditional Expectations. [8]

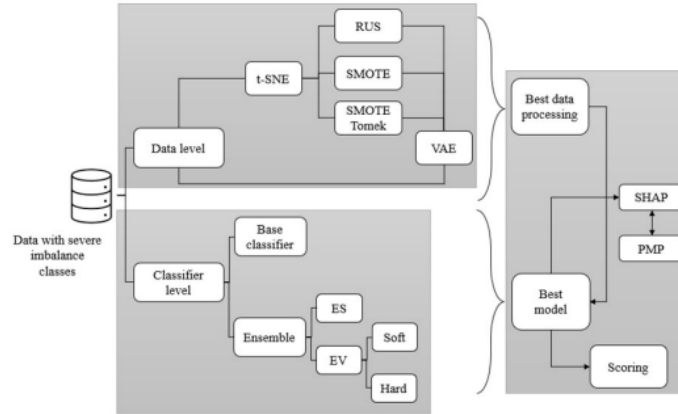


Fig. 1. SVEAD Interpretable framework: Anomaly Detection (Source: [8])

Another study proposed a framework that “incorporates innovative techniques designed to ensure representativeness, interpretability, and robustness”, to evaluate the resilience of Large Language Models to prompt injection attacks. 115 stimulated attacks based on coverage and relevance were chosen to ensure representativeness, and the responses generated from the attacks were assessed by a second LLM for increased interpretability. A confidence score and explanation were therefore produced, which is not possible by conventional malicious content classifiers. Finally, attacks with greater impact were assigned higher weights to compute a robust resilience score of the application. Results showed that newer models exhibit higher resilience than older models, which aligns with the prevailing notion. Additionally, to accommodate for emerging attack classifications and techniques, minimal adjustments were needed, which showcased the frameworks versatility. [3]

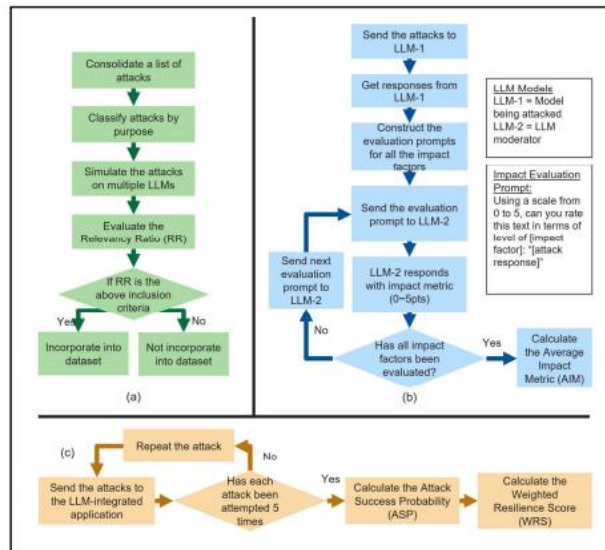


Fig. 1. The evaluation workflow [3]

In LLM performance interpretability, the role of Casual Inference has largely been overlooked by previous benchmarks that highlight the difference between generated and expected source code produced by LLMs in generative software tasks. However, accuracy is equally influenced by confounding variables, including lines of code, number of tokens, and prompt size. Therefore, one study proposed a benchmarking strategy, Galeras, to aid in interpreting LLM performance, which was composed of curated testbeds for code completion, summation, and generation. Results demonstrated that confounding bias can be reduced by casual inference evaluations, enabling an interpretable solution for analyzed accuracy metrics. [2]

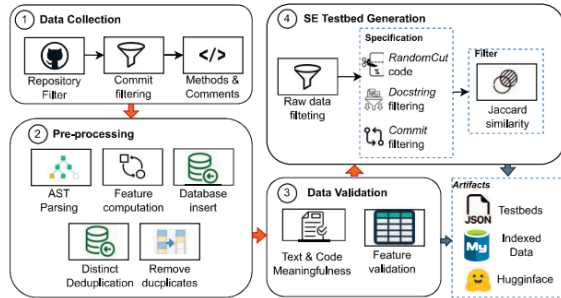


Fig. 1: Testbed Curation Pipeline of Galeras

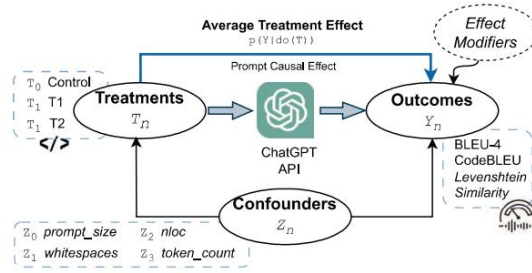


Fig. 2: Galeras Structural Causal Model Benchmarking [2]

xAI-Generative Adversarial Networks (xAI-GAN) was an architecture proposed in another study, in which xAI systems were used to modify the gradient descent process, to help overcome the resource and data-intensive training process shortcomings associated with GANs. xAI-GAN was demonstrated to help the discriminator be more easily fooled by the generator to produce "richer" corrective feedback regarding why the discriminator made its classification. xAI-GANs were shown to outperform GANs trained on Differentiable Augmentation (which helps make GANs data-efficient), and that superior results could be achieved by combining both techniques, giving users more control over the learning process of models. [9]

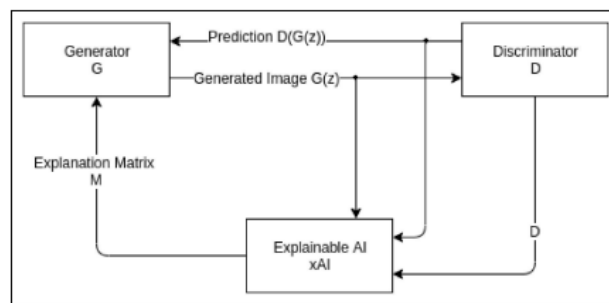
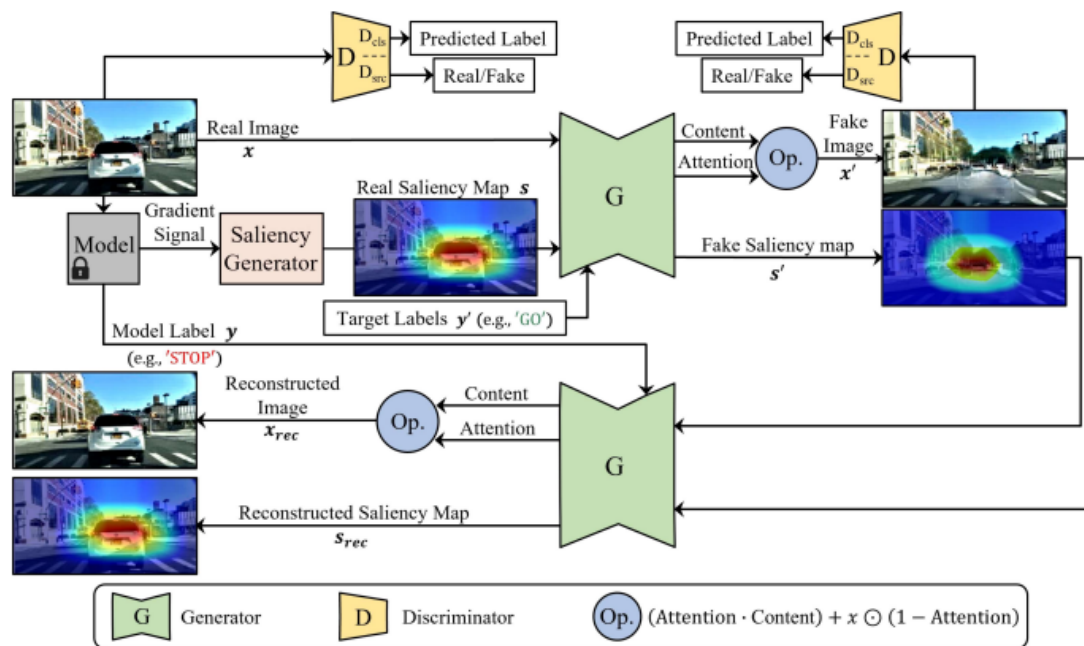


Figure 1: System Architecture of xAI-GAN. [9]

In safety-critical applications, the explainability of black-box deep neural networks is of utmost importance, and CounterFactual (CF) explanations, which compute the minimum input modifications required for the model's decision boundary to be crossed, have emerged to be a promising approach for interpretability. Unfortunately, instead of concentrating on DNNs

discriminative features, user-selected features that may not lie near the decision boundary are currently used in deep generative CF models, contradicting the objective of CFs. [10]



[10]

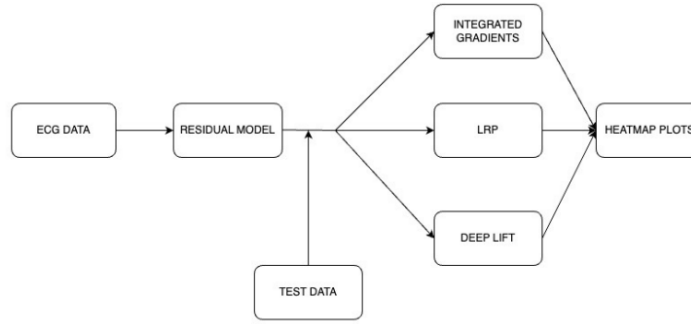


Fig. 2. Flowchart illustrating the XAI model for ECG analysis: Training the residual model with ECG data, testing with patient ECG signals, and employing explainable models (Integrated Gradients, LRP, and DeepLift) to generate prediction-concentrated heatmaps. [11]

Another study combined Markov logic networks (MLNs), for logical medical reasoning, with external knowledge extracted using LLMs and a search engine, for automatic knowledge acquisition and formulation, to improve interpretability and accuracy for explainable medical decision-making and produce transparent and trustworthy automated medical diagnostic systems. Specifically, a set of first-order logic (FOL) rules was generated which became a crucial component for the MLN-based diagnostic algorithm. This framework used neural methods for comprehensive patient information extraction, and interpretable symbolic reasoning for diagnosis, providing understandable reasoning for its decisions while maintaining the accuracy of deep neural networks. [12]

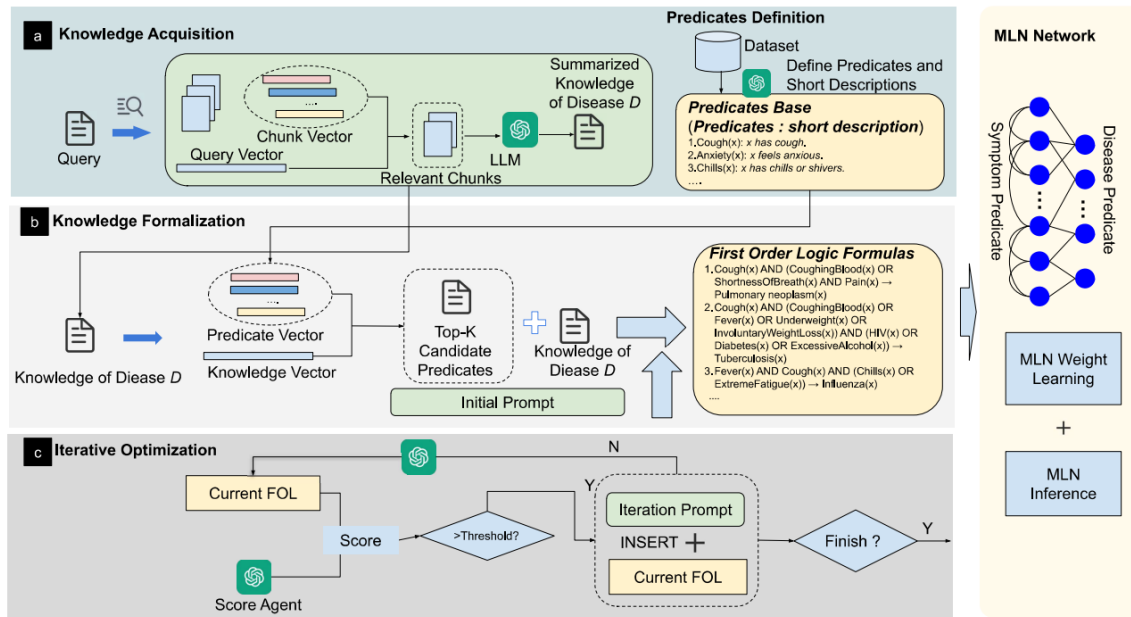


Fig. 1. The architecture of the proposed framework includes three critical steps: (a) Knowledge Acquisition, (b) Knowledge Formalization, and (c) Iterative Optimization. In particular, the Knowledge Acquisition step involves a search engine and LLMs to automatically collect and process critical domain knowledge. Then, the Knowledge Formalization step transforms the gathered domain knowledge into FOL using LLMs. Finally, the Iterative Optimization step enhances the quality of FOL through multiple iterations. After these three steps, our framework accumulates high-quality FOL rules of medical knowledge that will be used for the following MLN training and inference, resulting in an interpretable medical diagnosis system. [12]

In the economic industry, real-time tracking of economic indicators about public behaviors using social media data has previously relied on linear, simplistic models that provide limited exploratory power or insights into why these predictive methods work. Therefore, a study proposed Group Additive Gaussian Processes, a novel interpretable ML model based on Gaussian Process regression, to provide a robust non-parametric Bayesian learning framework and calibrated uncertainty measures with its predictions, and was shown to generalize well in scenarios with limited training data. [13]

Finally, in reinforcement learning, to both explain an agent's observations and interpret the relationships between agent states and actions, one study proposed a fuzzy centered explainable network (FCEN). FCEN established if-then rules to formulate state-action mappings with human-understandable logic, and a generative model to visualize learned knowledge, leveraging the interpretability of fuzzy neural networks. Experimental results proved high level performance in RL tasks, and both global explainability, achieved by fuzzy rule visualization, and local explainability, fulfilled by the attention mechanism of the CCT extractor and the FNN inner states, of RL agents' decision logic was boosted. [14]

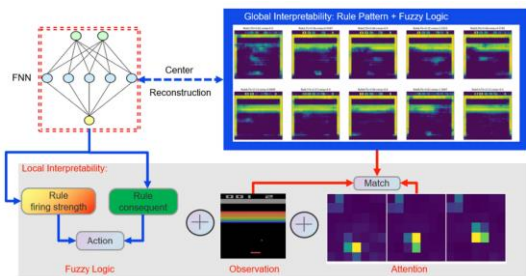


Fig. 1. Key components of the FCEN for explaining an agent's behavior. The FCEN maintains a global explanation through fuzzy rule pattern recognition and local explanations through the values of rule firing strengths and consequences.

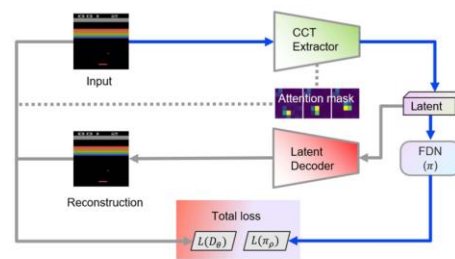


Fig. 2. FCEN architecture integrates and interacts with three components: CCT Extractor, FDN, and latent decoder. The CCT Extractor generates a latent vector from the input frame, which is used as input to both FDN and the latent decoder. FDN produces the final action of the agent, while the latent decoder help generates human-interpretable explanations of the decision-making process.

[14]

In conclusion, GenAI is exploding in popularity across the world, becoming implemented in a diverse range of industries and applications, and the possibilities for innovation seem endless. **However, the rapid explosion of GenAI does not come without risks** or potential drawbacks, due to the complex, black-box nature of GenAI that makes these models vulnerable to misgeneralization, hallucinations, biases, and adversarial attacks.

Therefore, when building GenAI models, interpretability and explainability is of utmost importance, to ensure these models are trustworthy, transparent, perform the way they are intended to, and have a high degree of resilience against deepfakes, disinformation, confidential data leakage, and other threats.

Several studies have investigated novel ways of trying to build resilient, trustworthy, and explainable GenAI models that don't compromise on performance, and have investigated their interpretability. However, no method has been widely deployed into practice, showcasing that there is still a lot of work and research to be conducted in this vital field.

Before the world fully embraces GenAI to help solve the most pressing issues plaguing the globe and increase the proficiency of humankind, widely accepted responsible practices and a global

consensus on the ethical usage of GenAI must be achieved, as well as proactive cyber-wellness programs to increase public awareness of the potential risks associated with GenAI to help safeguard citizens. This is especially the case for safety-critical applications and high reliability systems, where GenAI models that make incorrect predictions or are jailbroken could lead to catastrophic outcomes, but also have the potential to make revolutionary breakthroughs.

I therefore believe that going forward, it will be essential that standardized interpretability and explainability metrics are implemented and rigorously followed so that GenAI models can be deployed with confidence, and that the assessment criteria will be incorporated to differing degrees depending on the application industry and type of GenAI model.

References

- [1] K. Kenthapadi, H. Lakkaraju, and N. Rajani, "Generative AI meets Responsible AI: Practical Challenges and Opportunities". In Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '23). Association for Computing Machinery, New York, NY, USA, 2023, pp. 5805–5806. <https://doi.org/10.1145/3580305.3599557>
- [2] D. Rodriguez-Cardenas, D. N. Palacio, D. Khati, H. Burke and D. Poshyvanyk, "Benchmarking Causal Study to Interpret Large Language Models for Source Code," 2023 IEEE International Conference on Software Maintenance and Evolution (ICSME), Bogotá, Colombia, 2023, pp. 329-334, doi: 10.1109/ICSME58846.2023.00040.
- [3] D. W. Yip, A. Esmradi and C. F. Chan, "A Novel Evaluation Framework for Assessing Resilience Against Prompt Injection Attacks in Large Language Models," 2023 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE), Nadi, Fiji, 2023, pp. 1-5, doi: 10.1109/CSDE59766.2023.10487667.
- [4] M. R. Shoaib, Z. Wang, M. T. Ahvanooey and J. Zhao, "Deepfakes, Misinformation, and Disinformation in the Era of Frontier AI, Generative AI, and Large AI Models," 2023 International Conference on Computer and Applications (ICCA), Cairo, Egypt, 2023, pp. 1-7, doi: 10.1109/ICCA59364.2023.10401723.
- [5] V. Chamola, V. Hassija, A. R. Sulthana, D. Ghosh, D. Dhingra and B. Sikdar, "A Review of Trustworthy and Explainable Artificial Intelligence (XAI)," in IEEE Access, vol. 11, pp. 78994-79015, 2023, doi: 10.1109/ACCESS.2023.3294569.
- [6] O. Embarak, "Decoding the Black Box: A Comprehensive Review of Explainable Artificial Intelligence," 2023 9th International Conference on Information Technology Trends (ITT), Dubai, United Arab Emirates, 2023, pp. 108-113, doi: 10.1109/ITT59889.2023.10184238.

- [7] J. Yu et al., "INTERACTION: A Generative XAI Framework for Natural Language Inference Explanations," 2022 International Joint Conference on Neural Networks (IJCNN), Padua, Italy, 2022, pp. 1-8, doi: 10.1109/IJCNN55064.2022.9892336.
- [8] S. Maitra, V. Mishra, P. Verma, M. Chopra and P. Nath, "Sampling - Variational Auto Encoder - Ensemble: In the Quest of Explainable Artificial Intelligence," 2023 International Conference on Electrical and Information Technology (IEIT), Malang, Indonesia, 2023, pp. 43-50, doi: 10.1109/IEIT59852.2023.10335522.
- [9] V. Nagisetty, L. Graves, J. Scott, and V. Ganesh, "xAI-GAN: Enhancing Generative Adversarial Networks via Explainable AI Systems," arXiv.org, Mar. 29, 2022.
<https://arxiv.org/abs/2002.10438>
- [10] A. Samadi, A. Shirian, K. Koufos, K. Debattista and M. Dianati, "SAFE: Saliency-Aware Counterfactual Explanations for DNN-based Automated Driving Systems," 2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC), Bilbao, Spain, 2023, pp. 5655-5662, doi: 10.1109/ITSC57777.2023.10422102.
- [11] S. Khambampati, S. Dondapati, T. V. Kattamuri and R. K. Pathinarupothi, "CureNet: Improving Explainability of AI Diagnosis Using Custom Large Language Models," 2023 3rd International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON), Bangalore, India, 2023, pp. 1-7, doi: 10.1109/SMARTGENCON60755.2023.10442356.
- [12] H. Zhang, J. Li, Y. Wang and Y. Song, "Integrating Automated Knowledge Extraction with Large Language Models for Explainable Medical Decision-Making," 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Istanbul, Turkiye, 2023, pp. 1710-1717, doi: 10.1109/BIBM58861.2023.10385557.
- [13] N. Pokhriyal, B. Valentino, and S. Vosoughi, "An Interpretable Model for Real-time Tracking of Economic Indicators Using Social Media Data," ACM/IMS Transactions on Data Science, vol. 2, no. 4, pp. 1–32, Nov. 2021, doi: <https://doi.org/10.1145/3498332>.
- [14] L. Ou, Y. -C. Chang, Y. -K. Wang and C. -T. Lin, "Fuzzy Centered Explainable Network for Reinforcement Learning," in IEEE Transactions on Fuzzy Systems, vol. 32, no. 1, pp. 203-213, Jan. 2024, doi: 10.1109/TFUZZ.2023.3295055.